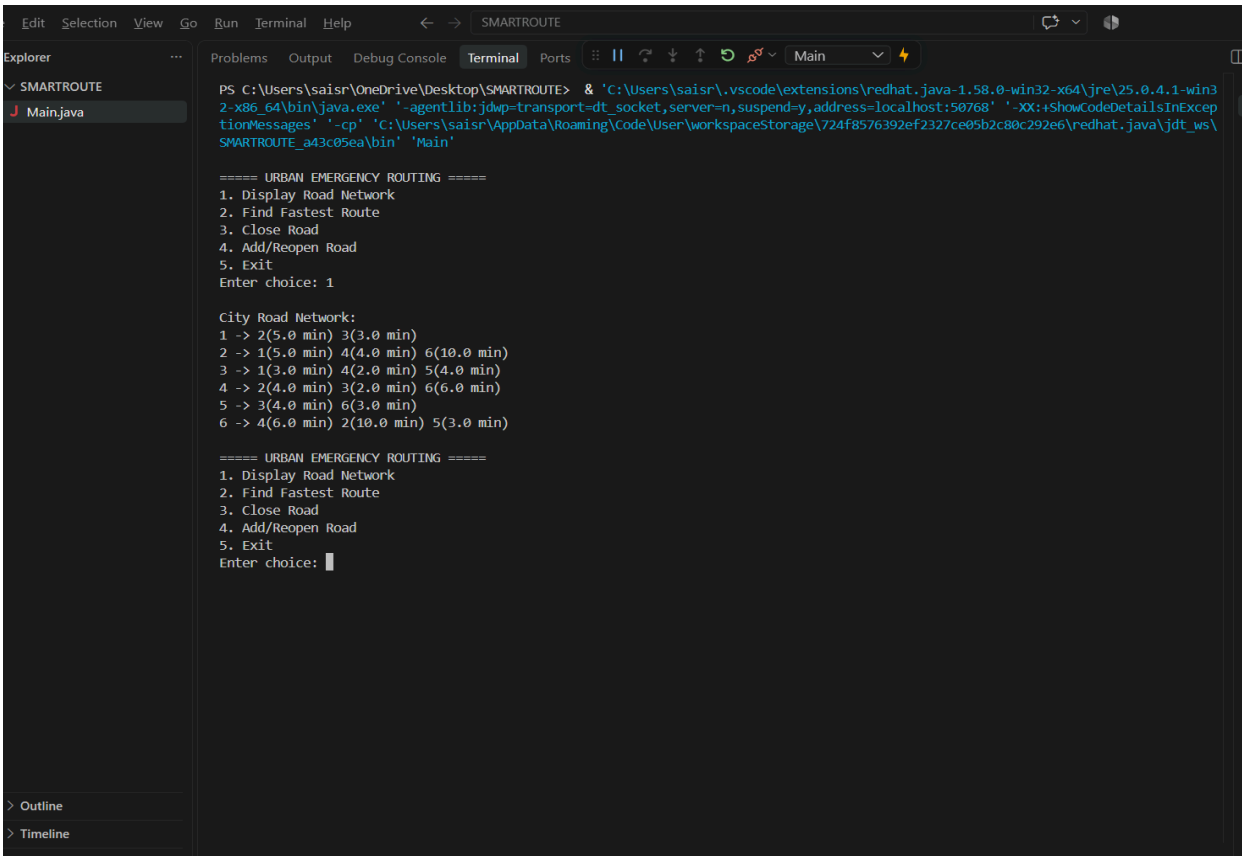
SMARTROUTE

Urban Emergency Road Routing System

Sample Execution Outputs and Test Case Results:

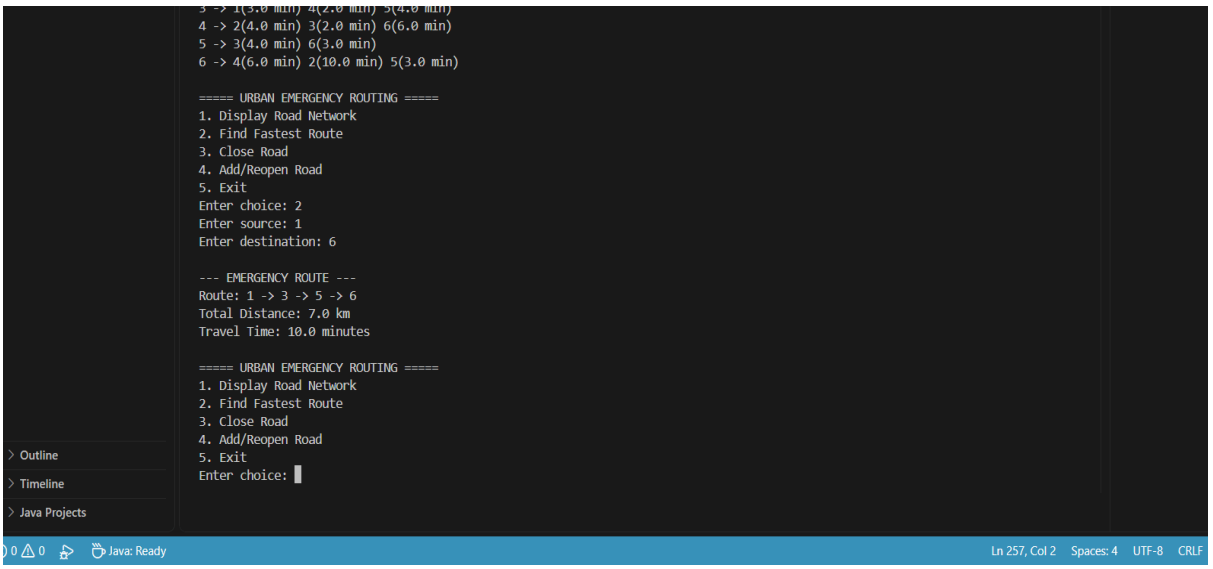
1. Test Case 1 - **Display City Road Network**

Objective: To display all available locations and connecting roads with their respective travel times.



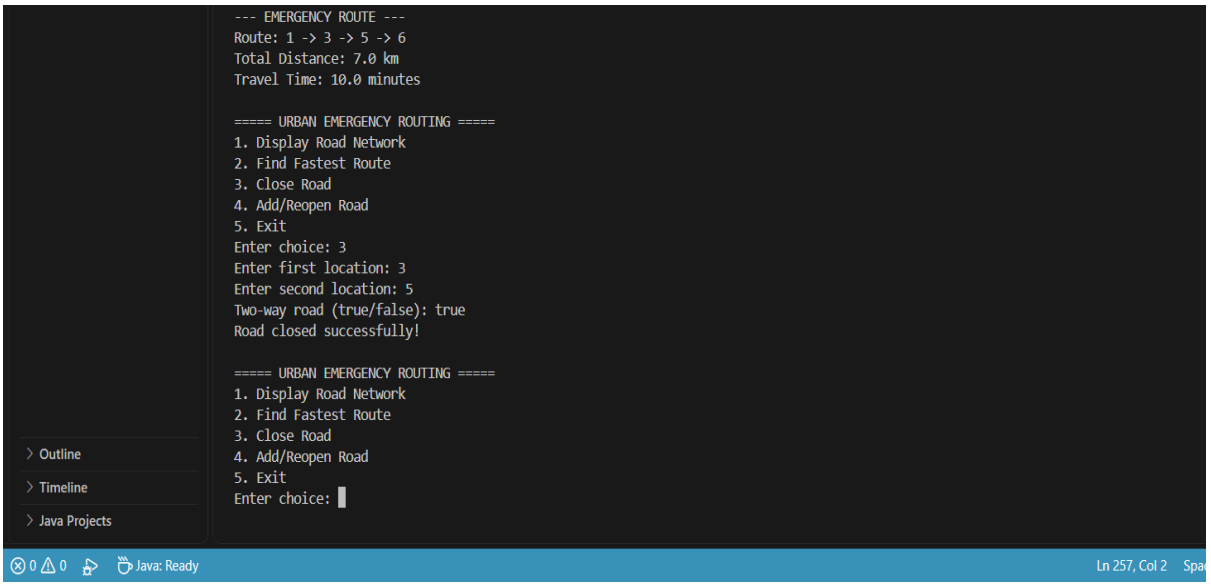
2. Test Case 2 - **Find Fastest Emergency Route**

Objective: To calculate the fastest route from the hospital to the accident location using Dijkstra's algorithm.



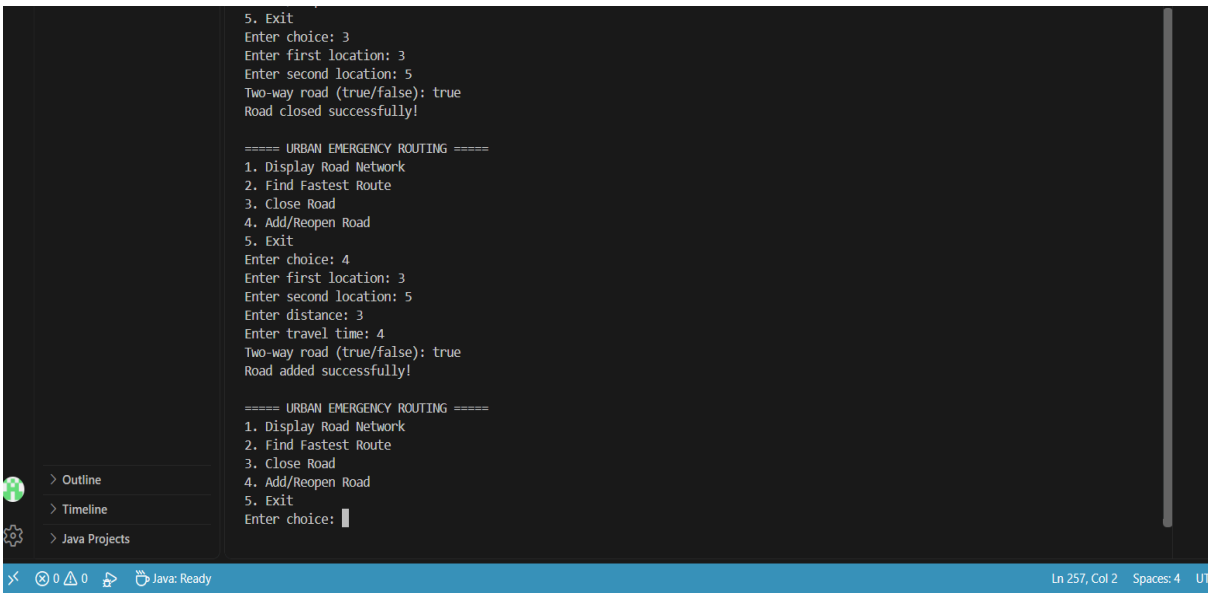
3. Test Case 3 - **Road Closure Simulation**

Objective: To simulate a road closure and remove the selected road from the city road network.



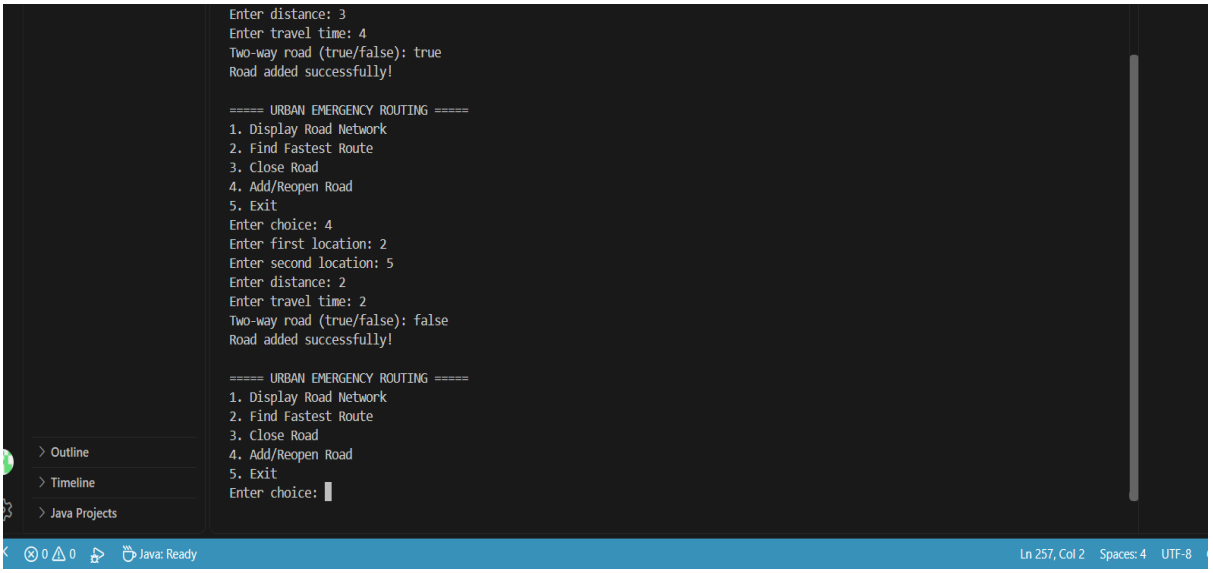
4. Test Case 4 - **Road Reopening**

Objective: To restore a previously closed road and update the available road network.



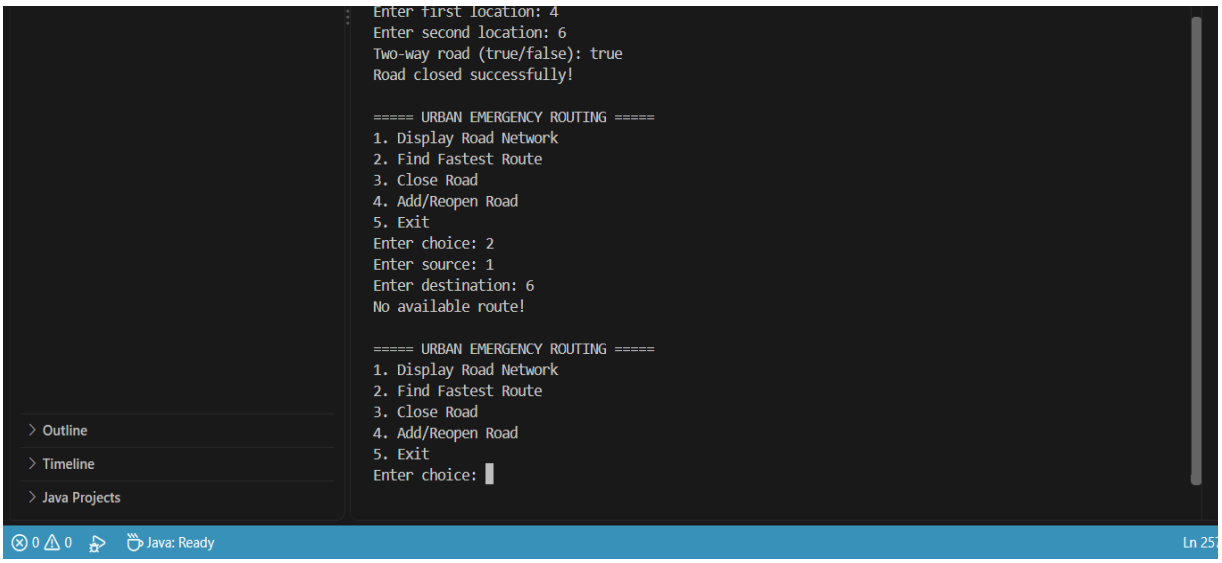
5. Test Case 5 - **One-Way Road**

Objective: Verify that the system respects one-way road restrictions.



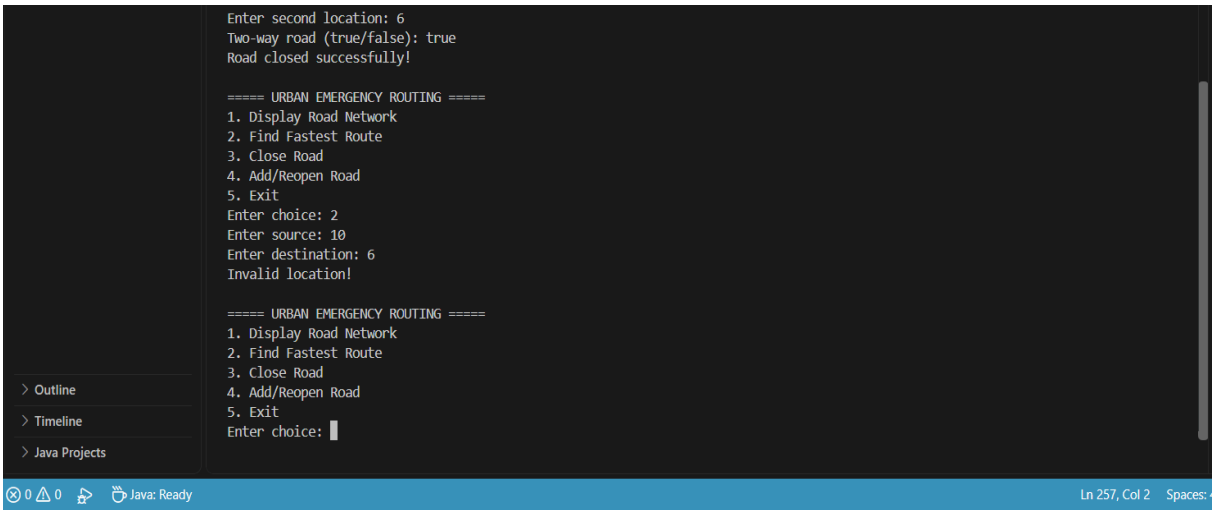
6. Test Case 6 - **No Available Road**

Objective: Test what happens when the destination cannot be reached.



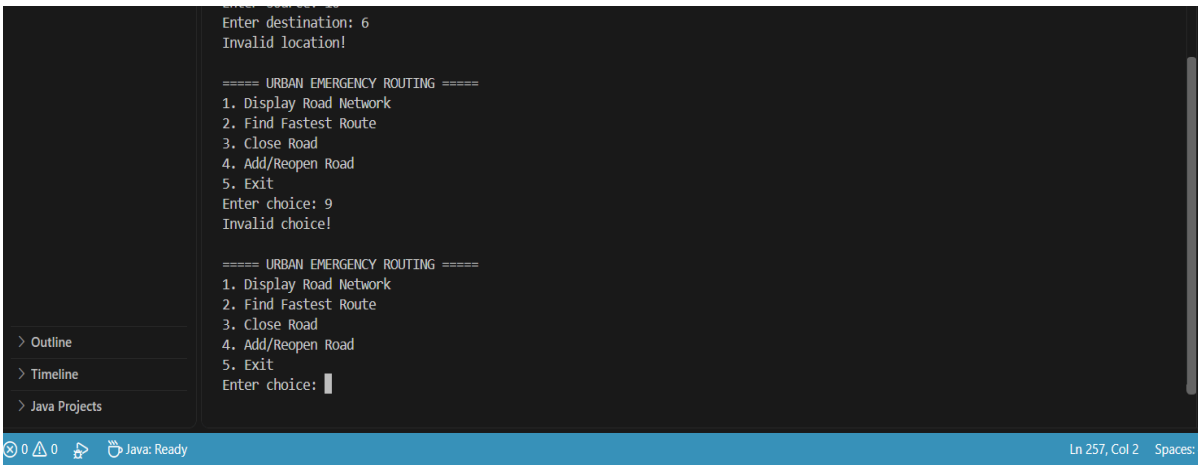
7. Test Case 7 - **Invalid Source or Destination**

Objective: Verify that the system handles invalid locations.



8. Test Case 8 - **Invalid Menu Choice**

Objective: Check whether the program handles incorrect menu selections.



9. Test Case 9 - **Exit the Program**

